

Laplace Transform

$$F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$$

**06016401**

## **MATHEMATICS FOR INFORMATION TECHNOLOGY**

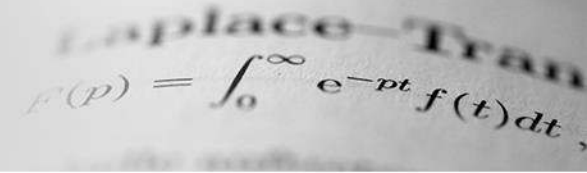
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# Outline

A decorative header image showing a close-up of a document with the Laplace Transform formula. The text "Laplace-Trans" is visible at the top, and the formula  $F(p) = \int_0^{\infty} e^{-pt} f(t) dt$  is written below it.
$$F(p) = \int_0^{\infty} e^{-pt} f(t) dt$$

- Matrices
  - The Inverse of a Matrix
- Determinants
  - The Determinant of a Matrix
  - Applications of Determinants
- Applications of Matrix Operations
  - Cryptography

# The Inverse of a Matrix

A decorative image in the top right corner showing a portion of a document with the Laplace Transform formula:  $F(p) = \int_0^{\infty} e^{-pt} f(t) dt$ .
$$F(p) = \int_0^{\infty} e^{-pt} f(t) dt$$

An  $n \times n$  matrix  $A$  is **invertible** (or **nonsingular**) if there exists an  $n \times n$  matrix  $B$  such that

$$AB = BA = I_n$$

where  $I_n$  is the identity matrix of order  $n$ . The matrix  $B$  is called the (multiplicative) **inverse** of  $A$ . A matrix that does not have an inverse is called **noninvertible** (or **singular**).

# Finding the Inverse of a Matrix

$$A^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Find the inverse of the matrix

$$A = \begin{bmatrix} 1 & 4 \\ -1 & -3 \end{bmatrix} \cdot \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\frac{1}{-3+4} \begin{bmatrix} -3 & -4 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} -3 & -4 \\ 1 & 1 \end{bmatrix}$$

# Properties of Inverse Matrices

A decorative image in the top right corner showing a close-up of a document with the Laplace Transform formula:  $F(p) = \int_0^{\infty} e^{-pt} f(t) dt$ .
$$F(p) = \int_0^{\infty} e^{-pt} f(t) dt$$

If  $A$  is an invertible matrix,  $k$  is a positive integer, and  $c$  is a scalar not equal to zero, then  $A^{-1}$ ,  $A^k$ ,  $cA$ , and  $A^T$  are invertible and the following are true.

1.  $(A^{-1})^{-1} = A$
2.  $(A^k)^{-1} = A^{-1}A^{-1} \cdot \cdot \cdot A^{-1} = (A^{-1})^k$
3.  $(cA)^{-1} = \frac{1}{c}A^{-1}, c \neq 0$
4.  $(A^T)^{-1} = (A^{-1})^T$

If  $A$  and  $B$  are invertible matrices of size  $n$ , then  $AB$  is invertible and

$$(AB)^{-1} = B^{-1}A^{-1}.$$

# The Determinant of a Matrix

- Every *square* matrix can be associated with a real number called its **determinant**.
- Historically, the use of determinants arose from the recognition of special patterns that occur in the solutions of systems of linear equations.

# Definition of the Determinant of a $2 \times 2$ Matrix

The **determinant** of the matrix

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

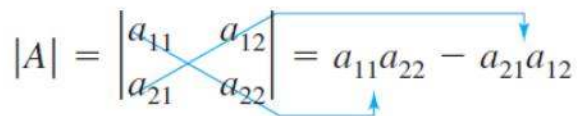
is given by

$$\det(A) = |A| = a_{11}a_{22} - a_{21}a_{12}.$$

**REMARK:** In this text,  $\det(A)$  and  $|A|$  are used interchangeably to represent the determinant of a matrix. Vertical bars are also used to denote the absolute value of a real number; the context will show which use is intended. Furthermore, it is common practice to delete the matrix brackets and write

$$\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} \quad \text{instead of} \quad \left| \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \right|.$$

A convenient method for remembering the formula for the determinant of a  $2 \times 2$  matrix is shown in the diagram below.

$$|A| = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{21}a_{12}$$


The determinant is the difference of the products of the two diagonals of the matrix. Note that the order is important, as demonstrated above.

# The Determinant of a Matrix of Order 2

Laplace-Trans  
 $F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$

Find the determinant of each matrix.

(a)  $A = \begin{bmatrix} 2 & -3 \\ 1 & 2 \end{bmatrix}$

$$4 + 3 = 7$$

(b)  $B = \begin{bmatrix} 2 & 1 \\ 4 & 2 \end{bmatrix}$

$$4 - 4 = 0$$

(c)  $C = \begin{bmatrix} 0 & 3 \\ 2 & 4 \end{bmatrix}$

$$0 - 6 = -6$$



# Definitions of Minors and Cofactors of a Matrix

$$\text{Laplace-Trans} \\ F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$$

If  $A$  is a square matrix, then the **minor**  $M_{ij}$  of the element  $a_{ij}$  is the determinant of the matrix obtained by deleting the  $i$ th row and  $j$ th column of  $A$ . The **cofactor**  $C_{ij}$  is given by

$$C_{ij} = (-1)^{i+j} M_{ij}.$$

For example, if  $A$  is a  $3 \times 3$  matrix, then the minors and cofactors of  $a_{21}$  and  $a_{22}$  are as shown in the diagram below.

*Minor of  $a_{21}$*

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}, \quad M_{21} = \begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix}$$

Delete row 2 and column 1.

*Cofactor of  $a_{21}$*

$$\begin{aligned} C_{21} &= (-1)^{2+1} M_{21} \\ &= -M_{21} \end{aligned}$$

*Minor of  $a_{22}$*

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}, \quad M_{22} = \begin{vmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{vmatrix}$$

Delete row 2 and column 2.

*Cofactor of  $a_{22}$*

$$\begin{aligned} C_{22} &= (-1)^{2+2} M_{22} \\ &= M_{22} \end{aligned}$$

# Definitions of Minors and Cofactors of a Matrix (cont.)

As you can see, the minors and cofactors of a matrix can differ only in sign. To obtain the cofactors of a matrix, first find the minors and then apply the checkerboard pattern of +’s and -’s shown below.

## Sign Pattern for Cofactors

$$\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}$$

$3 \times 3$  matrix

$$\begin{bmatrix} + & - & + & - \\ - & + & - & + \\ + & - & + & - \\ - & + & - & + \end{bmatrix}$$

$4 \times 4$  matrix

$$\begin{bmatrix} + & - & + & - & + & \dots \\ - & + & - & + & - & \dots \\ + & - & + & - & + & \dots \\ - & + & - & + & - & \dots \\ + & - & + & - & + & \dots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix}$$

$n \times n$  matrix

Note that *odd* positions (where  $i + j$  is odd) have negative signs, and even positions (where  $i + j$  is even) have positive signs.

# Find the Minors and Cofactors of a Matrix

Laplace-Trans  
 $F(p) = \int_0^{\infty} e^{-pt} f(t) dt$

Find all the minors and cofactors of

$$A = \begin{bmatrix} 0 & 2 & 1 \\ 3 & -1 & 2 \\ 4 & 0 & 1 \end{bmatrix}$$

$$M_{11} = \begin{bmatrix} -1 & 2 \\ 0 & 1 \end{bmatrix} = -1$$

$$M_{12} = \begin{bmatrix} 3 & 2 \\ 4 & 1 \end{bmatrix} = -5$$

$$M_{13} = \begin{bmatrix} 3 & -1 \\ 4 & 0 \end{bmatrix} = 4$$

$$M_{21} = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix} = 2$$

$$M_{22} = \begin{bmatrix} 0 & 1 \\ 4 & 1 \end{bmatrix} = -4$$

$$M_{23} = \begin{bmatrix} 0 & 2 \\ 4 & 0 \end{bmatrix} = -8$$

$$M_{31} = \begin{bmatrix} 2 & 1 \\ -1 & 2 \end{bmatrix} = 5$$

$$M_{32} = \begin{bmatrix} 0 & 1 \\ 3 & 2 \end{bmatrix} = -3$$

$$M_{33} = \begin{bmatrix} 0 & 2 \\ 3 & -1 \end{bmatrix} = -6$$

$$C_{11} = -1 - 0 = -1$$

$$C_{12} = -(3 - 8) = 5$$

$$C_{13} = 0 + 4 = 4$$

$$C_{21} = -(2 - 0) = -2$$

$$C_{22} = 0 - 4 = -4$$

$$C_{23} = -(0 - 8) = 8$$

$$C_{31} = 4 + 1 = 5$$

$$C_{32} = -(0 - 3) = 3$$

$$C_{33} = 0 - 6 = -6$$

# Definition of the Determinant of a Matrix



Laplace-Transform  
 $F(p) = \int_0^{\infty} e^{-pt} f(t) dt$

If  $A$  is a square matrix (of order 2 or greater), then the determinant of  $A$  is the sum of the entries in the first row of  $A$  multiplied by their cofactors. That is,

$$\det(A) = |A| = \sum_{j=1}^n a_{1j}C_{1j} = a_{11}C_{11} + a_{12}C_{12} + \cdots + a_{1n}C_{1n}.$$

# The Determinant of a Matrix of Order 3

Laplace-Trans  
 $F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$

Find the determinant of

$$A = \begin{bmatrix} 0 & 2 & 1 \\ 3 & -1 & 2 \\ 4 & 0 & 1 \end{bmatrix}.$$

6/8/2020  
Handwritten notes and arrows indicating the expansion process.

$$|A| = 0(-1) + 2(5) + 1(4)$$

$$= 14$$

$$|A| = 3(-2) + (-1)(-4) + 2(8)$$

$$= 14$$

$$|A| = 4(5) + 0(3) + 1(-6)$$

$$= 14$$

# Expansion by Cofactors



Laplace-Transform  
 $F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$

Let  $A$  be a square matrix of order  $n$ . Then the determinant of  $A$  is given by

$$\det(A) = |A| = \sum_{j=1}^n a_{ij}C_{ij} = a_{i1}C_{i1} + a_{i2}C_{i2} + \cdots + a_{in}C_{in}$$

or

$$\det(A) = |A| = \sum_{i=1}^n a_{ij}C_{ij} = a_{1j}C_{1j} + a_{2j}C_{2j} + \cdots + a_{nj}C_{nj}.$$

# The Determinant of a Matrix of Order 4

Laplace-Trans  
 $F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$

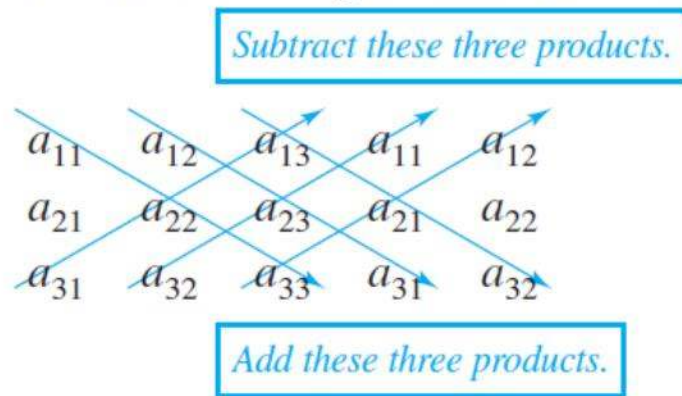
Find the determinant of

$$A = \begin{bmatrix} 1 & -2 & 3 & 0 \\ -1 & 1 & 0 & 2 \\ 0 & 2 & 0 & 3 \\ 3 & 4 & 0 & -2 \end{bmatrix}.$$



# Definition of the Determinant of a Matrix (cont.)

There is an alternative method commonly used for evaluating the determinant of a  $3 \times 3$  matrix  $A$ . To apply this method, copy the first and second columns of  $A$  to form fourth and fifth columns. The determinant of  $A$  is then obtained by adding (or subtracting) the products of the six diagonals, as shown in the following diagram.



Try confirming that the determinant of  $A$  is

$$|A| = a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{31}a_{22}a_{13} - a_{32}a_{23}a_{11} - a_{33}a_{21}a_{12}.$$



# The Determinant of a Matrix of Order 3

Laplace-Trans  
 $F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$

Find the determinant of

$$A = \begin{bmatrix} 0 & 2 & 1 \\ 3 & -1 & 2 \\ 4 & -4 & 1 \end{bmatrix} \begin{matrix} 0 & 2 \\ 3 & -1 \\ 4 & -4 \end{matrix}$$

$$= 0 + 16 - 12 + 4 - 0 - 6 = 2$$

# Conditions That Yield a Zero Determinant

Laplace-Trans  
 $F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$

If  $A$  is a square matrix and any one of the following conditions is true, then  $\det(A) = 0$ .

1. An entire row (or an entire column) consists of zeros.
2. Two rows (or columns) are equal.
3. One row (or column) is a multiple of another row (or column).

$$\begin{vmatrix} 0 & 0 & 0 \\ 2 & 4 & -5 \\ 3 & -5 & 2 \end{vmatrix} = 0,$$

The first row  
has all zeros.

$$\begin{vmatrix} 1 & -2 & 4 \\ 0 & 1 & 2 \\ 1 & -2 & 4 \end{vmatrix} = 0,$$

The first and third  
rows are the same.

$$\begin{vmatrix} 1 & 2 & -3 \\ 2 & -1 & -6 \\ -2 & 0 & 6 \end{vmatrix} = 0.$$

The third column is a  
multiple of the first column.

# Properties of Determinants

If  $A$  and  $B$  are square matrices of order  $n$ , then

$$\det(AB) = \det(A) \det(B).$$

If  $A$  is an  $n \times n$  matrix and  $c$  is a scalar, then the determinant of  $cA$  is given by

$$\det(cA) = c^n \det(A).$$

A square matrix  $A$  is invertible (nonsingular) if and only if

$$\det(A) \neq 0.$$

If  $A$  is invertible, then

$$\det(A^{-1}) = \frac{1}{\det(A)}.$$

If  $A$  is a square matrix, then

$$\det(A) = \det(A^T).$$

# The Adjoint of a Matrix

Laplace-Transform  
 $F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$

- If  $A$  is a square matrix, then the **matrix of cofactors** of  $A$  has the form

$$\begin{bmatrix} C_{11} & C_{12} & \cdots & C_{1n} \\ C_{21} & C_{22} & \cdots & C_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ C_{n1} & C_{n2} & \cdots & C_{nn} \end{bmatrix}.$$

- The transpose of this matrix is called the **adjoint** of  $A$  and is denoted by  $\text{adj}(A)$ . That is,

$$\text{adj}(A) = \begin{bmatrix} C_{11} & C_{21} & \cdots & C_{n1} \\ C_{12} & C_{22} & \cdots & C_{n2} \\ \vdots & \vdots & \ddots & \vdots \\ C_{1n} & C_{2n} & \cdots & C_{nn} \end{bmatrix}.$$

# Finding the Adjoint of a Square Matrix

Laplace-Transform  
 $F(p) = \int_0^{\infty} e^{-pt} f(t) dt$

Find the adjoint of

$$A = \begin{bmatrix} -1 & 3 & 2 \\ 0 & -2 & 1 \\ 1 & 0 & -2 \end{bmatrix} = \begin{bmatrix} 5 & 12 \\ 6 & 3 \\ 7 & 2 \end{bmatrix}$$

# The Inverse of a Matrix Given by Its Adjoint

Laplace-Trans  
 $F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$

If  $A$  is an  $n \times n$  invertible matrix, then

$$A^{-1} = \frac{1}{\det(A)} \text{adj}(A).$$

# Using the Adjoint of a Matrix to Find Its Inverse

Use the adjoint of

$$A = \begin{bmatrix} -1 & 3 & 2 \\ 0 & -2 & 1 \\ 1 & 0 & -2 \end{bmatrix}$$

to find  $A^{-1}$ .

$$\frac{1}{3} \cdot \begin{bmatrix} 6 & 6 & 7 \\ 1 & 0 & 1 \\ 2 & 3 & 2 \end{bmatrix} = \begin{bmatrix} \frac{4}{3} & 2 & \frac{7}{3} \\ \frac{1}{3} & 0 & \frac{1}{3} \\ \frac{2}{3} & 1 & \frac{2}{3} \end{bmatrix}$$

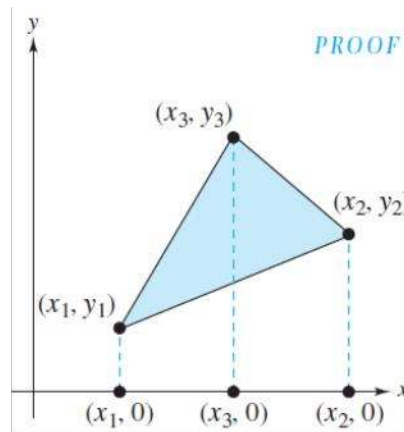
# Applications of Determinants

- Area, Volume, and Equations of Lines and Planes
- Two-Point Form of the Equation of a Line
- Volume of a Tetrahedron



# Area, Volume, and Equations of Lines and Planes

place-Trans  
 $F(p) = \int_0^\infty e^{-pt} f(t) dt,$



The area of the triangle whose vertices are  $(x_1, y_1)$ ,  $(x_2, y_2)$ , and  $(x_3, y_3)$  is given by

$$\text{Area} = \pm \frac{1}{2} \det \begin{bmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{bmatrix},$$

where the sign  $(\pm)$  is chosen to yield a positive area.

# Finding the Area of a Triangle

Find the area of the triangle whose vertices are  $(1, 0)$ ,  $(2, 2)$ , and  $(4, 3)$ .

$$\frac{1}{2} \begin{bmatrix} 1 & 0 & 1 \\ 2 & 2 & 1 \\ 4 & 3 & 1 \end{bmatrix} = -\frac{3}{2}$$

↓

$$\det = -3$$

# Two-Point Form of the Equation of a Line



Laplace-Trans  
 $F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$

An equation of the line passing through the distinct points  $(x_1, y_1)$  and  $(x_2, y_2)$  is given by

$$\det \begin{bmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{bmatrix} = 0.$$

# Finding an Equation of the Line Passing Through Two Points

Find an equation of the line passing through the points  $(2, 4)$  and  $(-1, 3)$ .

$$\begin{bmatrix} x & y & 1 \\ 2 & 4 & 1 \\ -1 & 3 & 1 \end{bmatrix} = 0$$

$$x \begin{bmatrix} 4 & 1 \\ 3 & 1 \end{bmatrix} - y \begin{bmatrix} 2 & 1 \\ -1 & 1 \end{bmatrix} + 1 \begin{bmatrix} 2 & 4 \\ -1 & 3 \end{bmatrix} = 0$$

$$x - 3y + 10 = 0$$

$$\text{Simplify} \Rightarrow x - 3y = -10$$

# Volume of a Tetrahedron

$$\text{Laplace-Trans} \\ F(p) = \int_0^\infty e^{-pt} f(t) dt,$$

- เตตระฮีดรอน ตรงกับภาษาอังกฤษว่า Tetrahedron (อ่านว่า tet-tra-he-dron) หมายถึงโครงสร้างปิระมิดฐานสามเหลี่ยม ซึ่งประกอบด้วย ด้าน (face) สามเหลี่ยม 4 ด้าน มุม (vertex) 4 มุม ขอบ (edge) 6 ขอบ เตตระฮีดรอนเป็นรูปทรงที่จัดอยู่ในประเภท พลาโตนิก (platonic)

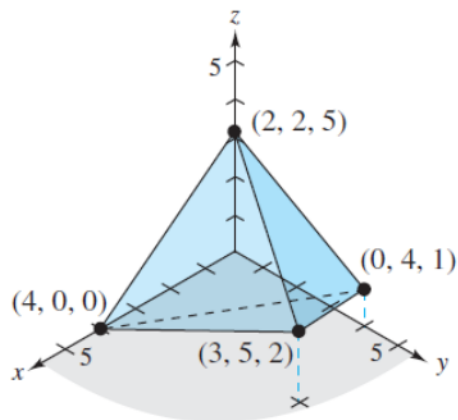
The volume of the tetrahedron whose vertices are  $(x_1, y_1, z_1)$ ,  $(x_2, y_2, z_2)$ ,  $(x_3, y_3, z_3)$ , and  $(x_4, y_4, z_4)$  is given by

$$\text{Volume} = \pm \frac{1}{6} \det \begin{bmatrix} x_1 & y_1 & z_1 & 1 \\ x_2 & y_2 & z_2 & 1 \\ x_3 & y_3 & z_3 & 1 \\ x_4 & y_4 & z_4 & 1 \end{bmatrix},$$

where the sign  $(\pm)$  is chosen to yield a positive volume.

# Finding the Volume of a Tetrahedron

Find the volume of the tetrahedron whose vertices are  $(0, 4, 1)$ ,  $(4, 0, 0)$ ,  $(3, 5, 2)$ , and  $(2, 2, 5)$



$$\frac{1}{6} \begin{bmatrix} 0 & 4 & 1 & 1 \\ 4 & 0 & 0 & 1 \\ 3 & 5 & 2 & 1 \\ 2 & 2 & 5 & 1 \end{bmatrix} = \frac{1}{6}(-72) = -12$$

$$\text{Volume} = 12$$

$$-4 \begin{bmatrix} 4 & 1 & 1 \\ 5 & 2 & 1 \\ 2 & 5 & 2 \end{bmatrix} + 1 \begin{bmatrix} 0 & 4 & 1 \\ 3 & 5 & 2 \\ 2 & 2 & 5 \end{bmatrix}$$

$$-24 - 48 = -72$$

$$\begin{array}{rrr} -4 & -20 & -5 \\ 4 & 1 & 1 \\ 5 & 2 & 1 \\ 2 & 5 & 2 \end{array}$$

$$+8 + 2 + 25$$

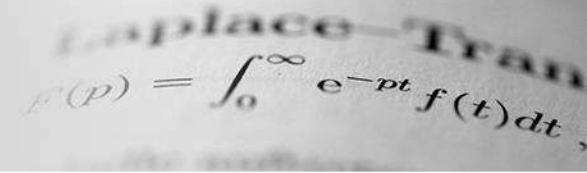
$$\begin{array}{rrr} -10 & 0 & -60 \\ 0 & 4 & 1 \\ 3 & 5 & 2 \\ 2 & 2 & 5 \end{array}$$

$$0 \quad 16 \quad 6$$

# Applications of Matrix Operations

- Cryptography

# Cryptography

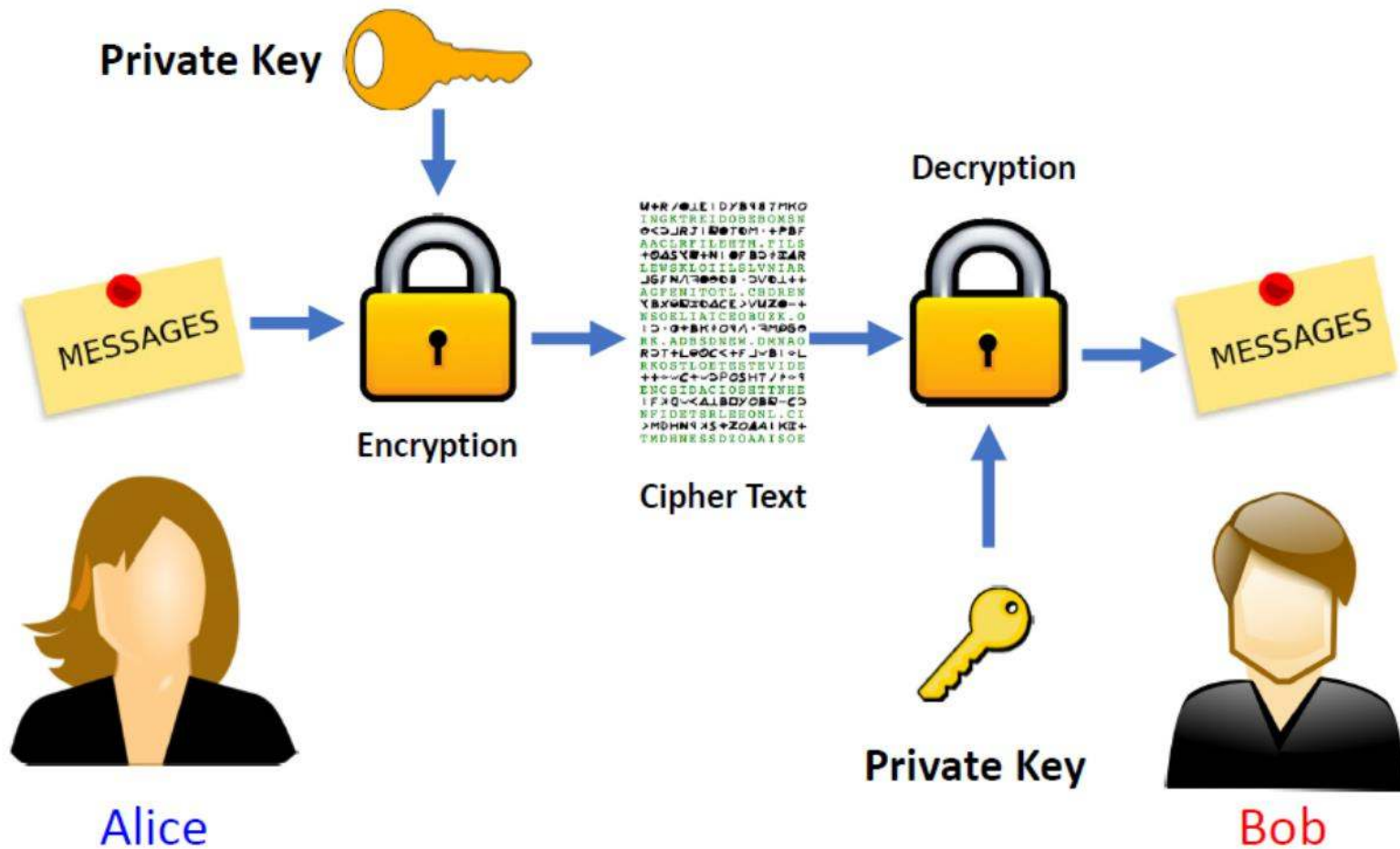
A decorative image in the top right corner showing a portion of a document with the Laplace Transform formula:  $F(p) = \int_0^{\infty} e^{-pt} f(t) dt$ . The text "Laplace-Trans" is visible above the formula.
$$F(p) = \int_0^{\infty} e^{-pt} f(t) dt$$

- A **cryptogram** is a message written according to a secret code (the Greek word *kryptos* means “hidden”).
- This section describes a method of using matrix multiplication to **encode** and **decode** messages.



# Symmetric Cryptography

Laplace-Trans  
$$F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$$



# Cryptography (cont.)

Laplace-Trans  
 $F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$

- Begin by assigning a number to each letter in the alphabet (with 0 assigned to a blank space), as follows.

|        |        |
|--------|--------|
| 0 = _  | 14 = N |
| 1 = A  | 15 = O |
| 2 = B  | 16 = P |
| 3 = C  | 17 = Q |
| 4 = D  | 18 = R |
| 5 = E  | 19 = S |
| 6 = F  | 20 = T |
| 7 = G  | 21 = U |
| 8 = H  | 22 = V |
| 9 = I  | 23 = W |
| 10 = J | 24 = X |
| 11 = K | 25 = Y |
| 12 = L | 26 = Z |
| 13 = M |        |

- Then the message is converted to numbers and partitioned into **uncoded row matrices**, each having  $n$  entries.

# Forming Uncoded Row Matrices

Laplace-Trans  
 $F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$

Write the uncoded row matrices of size  $1 \times 3$  for the message MEET ME MONDAY.

$$\begin{bmatrix} 13 & 5 & 5 \end{bmatrix} \begin{bmatrix} 20 & 0 & 13 \end{bmatrix} \begin{bmatrix} 5 & 0 & 13 \end{bmatrix} \begin{bmatrix} 15 & 11 & 4 \end{bmatrix} \begin{bmatrix} 1 & 25 & 0 \end{bmatrix}$$

# Cryptography (cont.)

A decorative image in the top right corner showing a portion of a document with the Laplace Transform formula:  $F(p) = \int_0^{\infty} e^{-pt} f(t) dt$ .

Laplace-Trans  
 $F(p) = \int_0^{\infty} e^{-pt} f(t) dt$

- To **encode** a message, choose an  $n \times n$  invertible matrix  $A$  and multiply the uncoded row matrices (on the right) by  $A$  to obtain **coded row matrices**.

# Encoding a Message



Laplace-Trans  
 $F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$

Use the matrix

$$A = \begin{bmatrix} 1 & -2 & 2 \\ -1 & 1 & 3 \\ 1 & -1 & -4 \end{bmatrix}$$

to encode the message MEET ME MONDAY.

# Cryptography (cont.)

Laplace-Trans  
 $F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$

- For those who do not know the matrix  $A$ , decoding the cryptogram found in the previous example is difficult.
- But for an authorized receiver who knows the matrix  $A$ , decoding is simple.
- The receiver need only multiply the coded row matrices by  $A^{-1}$  to retrieve the uncoded row matrices.
- In other words, if

$$X = [x_1 \ x_2 \ \cdots \ x_n]$$

is an uncoded  $1 \times n$  matrix, then  $Y = XA$  is the corresponding encoded matrix.

- The receiver of the encoded matrix can decode  $Y$  by multiplying on the right by  $A^{-1}$  to obtain

$$YA^{-1} = (XA)A^{-1} = X.$$

# Decoding a Message

Laplace-Trans  
 $F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$

Use the inverse of the matrix

$$A = \begin{bmatrix} 1 & -2 & 2 \\ -1 & 1 & 3 \\ 1 & -1 & -4 \end{bmatrix}$$

to decode the cryptogram

13 -26 21 33 -53 -12 18 -23 -42 5 -20 56 -24 23 77.

$$A^{-1} = \frac{1}{\det(A)} \operatorname{Adj}(A) = (1) \begin{bmatrix} -1 & -1 & 0 \\ -10 & -6 & -1 \\ -8 & -5 & -1 \end{bmatrix}$$

# Decoding a Message (cont.)



# Q & A

A close-up photograph of a document showing the Laplace transform formula. The text "Laplace-Trans" is visible at the top, followed by the equation  $F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$  and the phrase "with initial conditions" is partially visible below it.

Laplace-Trans  
 $F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$   
with initial conditions